

Intelligent AI Customer Care Assistant for Hanoi Heart Hospital

A. Background

Hanoi Heart Hospital is a Grade I cardiovascular specialty hospital and one of Vietnam's leading referral centers for cardiac care. The hospital serves approximately **2,500–3,000 outpatients per day**, creating a significant demand for information from patients and their families.

The most common inquiries include:

- Appointment booking
- Doctors' schedules
- Medical examination and treatment procedures
- Health insurance (BHYT) benefits
- Service pricing
- Hospital admission procedures
- Follow-up appointment guidance
- Information about the hospital's specialized medical services

Currently, these inquiries are handled through the customer service hotline, the hospital website, social media channels, and reception staff. The large volume of repetitive questions places considerable pressure on customer service personnel, resulting in delayed responses and inconsistent patient experiences.

B. Challenge

Develop an **AI Customer Care Assistant** that can be directly integrated into the **Hanoi Heart Hospital website** to assist patients and their families in accessing hospital information and answering frequently asked questions.

The solution should, at a minimum, satisfy the following requirements:

1. Knowledge-based Question Answering

Accurately answer questions related to:

- Appointment booking
- Medical examination and treatment procedures



- Health insurance (BHYT) benefits
- Medical service pricing
- Hospital working hours
- Doctors and medical departments
- Other official hospital information

2. Hospital System Integration

Support integration with the hospital's APIs or information systems to:

- Retrieve appointment schedules
- Redirect users to appointment booking services (Website, Zalo Mini App, or Customer Service Hotline)
- Retrieve service information when available

3. Conversational Experience

- Support **text-based conversations**
- Bonus for supporting **Vietnamese speech recognition (ASR)** and **speech synthesis (TTS)**

4. Trustworthy AI Responses

All responses **must be grounded in the hospital's official knowledge base**.

The AI **must not hallucinate or fabricate information**. If sufficient information is unavailable, the assistant should clearly state this and direct users to the appropriate hospital support channels.

5. Emergency Handling

When detecting symptoms that may indicate a medical emergency (e.g., severe chest pain, shortness of breath, fainting), the AI **must not provide treatment advice**.

Instead, it should immediately instruct users to seek emergency medical care or visit the hospital's Emergency Department according to the hospital's official procedures.

6. Deployment Readiness



The solution should be deployable on the hospital's infrastructure and comply with applicable requirements for:

- Data privacy
- Information security
- Healthcare data protection regulations

